People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
BADJI Mokhtar University Annaba
Faculty of Medicine
General Anatomy Laboratory

Foot joints & Talocrural joint

Prof. A. ABDALLAH
General Anatomy Laboratory
Faculty of Medicine – Annaba
Email: abdeldjalil.abdallah@univ-annaba.dz

Course outline

- **Introduction: "The joints of the foot"**
- **Talocrural joint :**
  - **Articular surfaces**
  - **Means of union**
  - **Synovial**
  - **Functional Anatomy**

Introduction: "The joints of the foot"

The joints of the foot are divided into three classes:

a. **Talocrural joint :**
   It is a diarthrosis, of the ginglyme type, it connects the two bones of the leg to the talus.
   It is a recessed joint, ensuring the stability of the foot during movements (walking and running).

b. **Tarsal joints :**
   They include:
   - Subtalar joint;
   - Transverse joint of the foot;
   - Articulation of the distal tarsal bones with each other.

c. **Forefoot joints :**
   They include:
   - Tarsometatarsal joints ;
   - Intermetatarsals joints ;
   - Metatarsophalangeal joints ;
   - Interphalangeal joints of the foot.

All these joints allow the movements of the foot: flexion, extension, adduction, abduction, supination, pronation, inversion and eversion.

Talocrural joint

1. Articular surfaces

There are two of them:

a. Tibiofibular articular surfaces :

It is made up of:

- Inferior articular surface of the tibia
- Articular facet of the medial malleolus
- Articular facet of the lateral malleolus

b. Articular surfaces of the trochlea of the talus:

It is a large, protruding joint. It has three surfaces:

- Upper surface of the trochlea: articular with the lower surface of the tibia;
- medial malleolar surface: comma-shaped, articular with the medial malleolus;
- Lateral malleolar surface: triangular, articular with the lateral malleolus;

How are all these articular surfaces organized ? The tibiofibular surfaces form a mortise into which the trochlea of the talus fits.

2. Means of union

a. Joint capsule:

It attaches to the periphery of the joint surfaces.

b. Passive ligaments :

- **Medial ligament:**

  Also called the deltoid ligament. It is strong and triangular, its apex attaching to the medial malleolus. It consists of two layers:

  Surface layer: includes from front to back:
  - Tibionavicular bundle
  - Tibio-calcaneal bundle

  Deep layer: includes from front to back:
  - Anterior tibiotalar bundle
  - Posterior tibiotalar bundle

- **Lateral ligament:**

  It extends from the lateral malleolus and terminates in three bundles:

  Anterior talofibular bundle
  Calcaneofibular bundle
  Posterior talofibular bundle

  Noticed: The lateral ligament is known as "the ankle sprain ligament".

3. Synovial

It lines the inner surface of the capsule.
It forms a cul-de-sac extension between the tibia and the fibula.

4. Functional Anatomy

The talocrural joint has only one degree of freedom, allowing flexion (or dorsiflexion, or elevation) and extension (or plantar flexion, or lowering) of the foot.
These movements occur around a transverse axis.

- Flexion: brings the dorsal side of the foot to the anterior side of the leg, his amplitude is from 20° to 30°.
- Extension: moves the dorsal surface of the foot away from the anterior surface of the leg, his amplitude is from 30° to 60°.